

Deep Learning

Deep Learning is a branch of artificial intelligence (AI) and machine learning that uses artificial neural networks with multiple layers to learn patterns from large amounts of data. Inspired by the structure of the human brain, these networks process information through interconnected nodes called neurons. Deep learning has become one of the most influential technologies in modern computing because it can automatically discover complex relationships in data without requiring extensive manual feature engineering.

Deep learning models are trained using large datasets and optimization algorithms such as gradient descent. During training, the network adjusts its internal parameters (weights) to minimize prediction errors. Common architectures include feedforward neural networks, convolutional neural networks (CNNs) for image processing, recurrent neural networks (RNNs) for sequential data, and transformer models that power many modern language and vision systems.

Applications of deep learning include image recognition, speech processing, natural language understanding, recommendation systems, medical diagnosis, autonomous vehicles, and generative AI. Its strengths lie in handling unstructured data such as text, audio, and images. However, deep learning often requires significant computational resources, large datasets, and careful model design to achieve optimal results.

As computing power and data availability continue to grow, deep learning remains a driving force behind advances in AI, enabling systems to perform tasks that were once considered uniquely human.